ModelIt!

Teacher's Guide

How Fast Does Ice Melt?

Exploring Reversible Changes in Matter

Quick Reference

Grade Level	Grade 2
NGSS Standard	2-PS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 2-PS1-4

Construct an argument with evidence that some changes caused by heating or cooling can be reversed and some cannot.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Engaging in Argument from Evidence** — Students construct an argument with evidence about what affects ice melting speed and whether the change can be reversed.
Disciplinary Core Idea	**PS1.B Chemical Reactions** — Heating or cooling a substance may cause changes that can be observed. Sometimes these changes are reversible (like melting/freezing) and sometimes they are not.
Crosscutting Concept	**Cause and Effect** — Students identify temperature as the cause of melting speed changes and gather evidence to support their explanations.

CAST Testing Alignment

MULTIPLE CHOICE:

Two ice cubes of the same size are placed outside. One is in the shade and one is in direct sunlight. Which ice cube will melt first and why?

CONSTRUCTED RESPONSE:

Explain what happens when an ice cube melts. Is this change reversible? Use evidence from your investigation to support your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every winter, icicles form and melt, ponds freeze and thaw, and snowflakes turn to puddles. These everyday events demonstrate one of science's most important ideas: matter can change form, and some changes can be reversed.

How Melting Works

Ice is solid water — its tiny molecules are locked together in a rigid structure. When heat energy flows into the ice from warmer surroundings, the molecules start vibrating faster. At 0 degrees Celsius (32 degrees Fahrenheit), the molecules break free of their rigid bonds and start sliding past each other. The solid ice becomes liquid water. The warmer the surroundings, the faster heat transfers into the ice, and the faster it melts.

Reversible vs. Irreversible Changes

Melting and freezing are reversible changes. Water can go from liquid to solid and back again unlimited times without changing what it is — it is always water. Other changes are irreversible. When you bake a cake, you cannot un-bake it. When you burn wood, you cannot un-burn it. The key difference is that reversible changes only change the FORM of the matter, not WHAT it is.

Insulators and Heat Transfer

Some materials slow down the transfer of heat. These are called insulators. Thick fabric, foam, and trapped air all make good insulators. This is why a cooler keeps ice from melting — the insulated walls slow down heat from the warm outside air reaching the cold ice inside. It is also why you wear a coat in winter — the coat insulates your body heat and keeps the cold air out.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify surrounding temperature as an external component and melting speed and ice size as internal components that change based on how warm or cool the environment is.

E **Establish Relationships**
Students determine that higher surrounding temperature causes faster melting speed (positive), and faster melting speed causes ice size to shrink more quickly (negative).

V **Visualize & Model**
Students build a model showing how the surrounding temperature drives melting speed and controls how quickly ice disappears.

E **Evaluate Outcomes**
Students run warm room, cold spot, and hot sunshine scenarios to compare melting speeds under different temperature conditions.

R **Revise & Extend**
Students add insulation or ice surface area to explore other factors that affect how fast ice melts.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Who has ever held an ice cube and watched it melt in your hand? What happened? Where did the ice go?"*

Watch For

Give each table group an ice cube on a plate. Ask: What is happening to it right now?

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to investigate what makes ice melt fast or slow — and whether we can reverse it!"*

Watch For

Read objectives together. Introduce melting, freezing, reversible change, and temperature.

Slide 3: The Big Question

3 min

Say: *"Does ice always melt at the same speed? Or can we make it melt faster or slower? What do you think controls it?"*

Watch For

Quick brainstorm: What could make ice melt FAST? What could keep it frozen LONG? List ideas on the board.

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a model to figure out what controls how fast ice melts."*

Watch For

Preview LEVER steps. Show ice cubes in two spots — sunny windowsill and shady corner.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What affects ice melting? The temperature around the ice is our input. Melting speed and ice size are what we observe."*

Watch For

Guide sorting: Surrounding Temperature is external; Melting Speed and Ice Size are internal.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When the temperature goes UP, does melting go faster or slower? And what happens to the ice size?"*

Watch For

Draw positive arrow from Temperature to Melting Speed. Draw negative arrow from Melting Speed to Ice Size (ice shrinks).

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Let us test! One ice cube in a warm spot, one in a cold spot. Predict which melts first!"*

Watch For

Students predict, then run simulations or observe real ice cubes. Record melting times. Compare results.

Slide 8: What Did We Discover?

7 min

Say: *"The best part? We can REVERSE this! If we freeze the melted water, it becomes ice again. Not all changes can be undone — but this one can!"*

Watch For

Pour melted ice water into a cup to put in the freezer. Compare to baking a cookie (irreversible). Discuss the difference.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Surrounding Temperature
- **INTERNAL:** Melting Speed, Ice Size

Relationships

- **Surrounding Temperature to Melting Speed:** POSITIVE (+) — Warmer surroundings transfer more heat energy to the ice, making it melt faster. On a hot day, ice melts very quickly. In a refrigerator, ice melts extremely slowly because there is very little heat to transfer.
- **Melting Speed to Ice Size:** NEGATIVE (-) — When the ice melts faster, it gets smaller more quickly. The solid ice is turning into liquid water, so the ice cube shrinks. Faster melting means the ice disappears sooner.

Key Discoveries

- Ice melts faster in warmer temperatures because more heat energy transfers to the ice
- In cold temperatures, ice melts very slowly or not at all because there is less heat to transfer
- Melting is a reversible change — you can freeze the water back into ice by cooling it down
- The amount of water from a melted ice cube is the same amount of matter as the original ice cube

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Ice melts because it wants to be water"	Ice does not 'want' anything — it is not alive! Ice melts because heat energy from the warmer surroundings flows into the ice. When enough heat energy enters, the solid structure breaks apart and the ice becomes liquid water. It is all about energy, not choice.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The water from melted ice is less than the original ice"	The melted water and the original ice are the same amount of matter! Nothing disappears during melting — the solid just changes into a liquid. If you freeze the water back into ice, you get the same amount of ice back.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"You cannot freeze water back into ice at home"	You absolutely can! Just pour water into an ice tray and put it in the freezer. In a few hours, it will be solid ice again. This is what makes melting a REVERSIBLE change — you can go back and forth between solid and liquid over and over.